## Developer Agent

> **PRIMARY ROLE:** Code Generation & Implementation
> **TECHNOLOGY:** Integrated in Timer Function, Azure OpenAI, LibGit2Sharp

✓ **AI-Powered Code Generation**

- GetCodeChangesAsync: AI generates code from work items
- Multimodal input processing (text + images)
- Context-aware generation using AST summary
- Adapts strategy for new/existing projects
- Token optimization (max 8000 tokens, temperature 0.1)

✓ **New Project Creation**

- Complete .NET 8 solution scaffolding
- Backend (API) + Frontend (React) structure
- Clean Architecture patterns
- Enterprise-grade standards:
  - Dependency Injection setup
  - Serilog structured logging
  - Health checks and monitoring
  - Authentication & Authorization (JWT/OAuth ready)
  - Swagger/OpenAPI documentation
  - Entity Framework Core with migrations
  - CORS configuration
  - Docker containerization

✓ **Existing Project Enhancement**

- Context-aware modifications
- Respects existing architecture
- Maintains code consistency
- Follows established patterns
- Integrates with existing components

✓ **Code Quality**

- Follows enterprise patterns
- Clean Code principles
- SOLID principles
- DRY (Don't Repeat Yourself)
- Comprehensive error handling
- Proper logging integration

✓ **Git Operations (LibGit2Sharp)**

- Clone repositories
- Create feature branches: feature/ai/{JiraKey}
- Commit with meaningful context messages
- Push to remote repository
- Handle authentication

✓ **Azure DevOps Integration**

- Create Pull Requests automatically
- Link PRs to Jira work items
- Add PR descriptions with context
- Set appropriate reviewers

✓ **Documentation Generation**

- Create Confluence architecture documentation
- Extract architectural decisions
- Document API endpoints
- Include implementation notes
- Link to Jira work items

✓ **Cleanup & Error Handling**

- Robust cleanup with retry logic
- Remove read-only attributes
- Handle file system locks
- Cleanup temporary directories
- Comprehensive error logging

**IMPLEMENTATION DETAILS:**

- Services: AIService, GitService, ConfluenceService
- AI Model: Azure OpenAI GPT-4
- Git Library: LibGit2Sharp
- Response Format: JSON with file changes

**CODE GENERATION STRUCTURE:**

```
For New Projects:
 ProjectName/
  ├── ProjectName.sln
  ├── BE/
  │   ├── ProjectName.Api/       (Web API)
  │   │   ├── Controllers/       (REST endpoints)
  │   │   ├── Models/            (Data models)
  │   │   ├── Services/          (Business logic)
  │   │   └── Program.cs         (Entry point)
  │   └── ProjectName.Tests/     (Unit tests)
```

```
├── FE/
│   └── ProjectName.Web/       (React SPA)
│       ├── src/components/    (UI components)
│       ├── src/services/      (API clients)
│       └── package.json
└── docs/
    ├── README.md
    └── API.md
```

**WORKFLOW:**

1. Receive work item + image attachments
2. Receive AST solution analysis
3. Determine project mode (new/existing)
4. Build appropriate prompt (new/enhancement)
5. Add multimodal context (images)
6. Call AI GetCodeChangesAsync
7. Parse JSON code changes response
8. Validate generated code
9. Apply code changes to repository
10. Extract architecture information
11. Create Confluence documentation
12. Git commit with context message
13. Push to feature branch
14. Create Azure DevOps Pull Request
15. Remove read-only attributes
16. Cleanup temporary directory

**Developer Agent Work Flow Diagram**

```
PHASE 4: CODE IMPLEMENTATION
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┌──────────────────────────────────┐
│   DEVELOPER AGENT                │
│   (Integrated in Timer Function) │
│                                  │
│   ┌──────────────────────────┐   │
│   │ Receives Validated Plan + │   │
│   │ Images from Scrum Master  │   │
│   └──────────────────────────┘   │
│            │                     │
│            ▼                     │
│   ┌──────────────────────────┐   │
│   │ Detect Project Type:      │   │
│   │ • New Project             │   │
│   │ • Existing Enhancement    │   │
│   └──────────────────────────┘   │
│            │                     │
│            ▼                     │
│   ┌──────────────────────────┐   │    ┌──────────────────┐
│   │ Build AI Prompt with:     │   │    │                  │
│   │ • AST Summary             │   │    │  Azure OpenAI    │
│   │ • Work Item Description   │──────►│  (O3 Mini/GPT 4)  │
```

```
25 |    | • Image Context           |  |     |  Multimodal      |
26 |    | • Project Type            |  |     |                  |
27 |    └───────────────────────────┘  |     └──────────────────┘
28 |            |                      |             |
29 |            |                      |             |
30 |            ▼                      |             |
31 |    ┌───────────────────────────┐  |    Generated Code
32 |    | Parse JSON Response:      |◄┘
33 |    | • File paths              |  |
34 |    | • File contents           |  |
35 |    | • Architecture info       |  |
36 |    └───────────────────────────┘  |
37 |            |                      |
38 |            ▼                      |
39 |    ┌───────────────────────────┐  |     ┌──────────────────┐
40 |    | Apply Code Changes to     |──────►| Local Repository |
41 |    | Repository                |  |     └──────────────────┘
42 |    └───────────────────────────┘  |
43 |            |                      |
44 |            ▼                      |
45 |    ┌───────────────────────────┐  |     ┌──────────────────┐
46 |    | Git Operations:           |  |     |                  |
47 |    | • Create feature branch   |──────►| Git Remote       |
48 |    | • Commit changes          |  |     | (Azure DevOps)   |
49 |    | • Push to remote          |  |     |                  |
50 |    └───────────────────────────┘  |     └──────────────────┘
51 |            |                      |
52 |            ▼                      |
53 |    ┌───────────────────────────┐  |     ┌──────────────────┐
54 |    | Create Pull Request       |──────►| Azure DevOps API |
55 |    └───────────────────────────┘  |     └──────────────────┘
56 |            |                      |
57 |            ▼                      |
58 |    ┌───────────────────────────┐  |     ┌──────────────────┐
59 |    | Generate Architecture     |──────►| Confluence API   |
60 |    | Documentation             |  |     └──────────────────┘
61 |    └───────────────────────────┘  |
62 |            |                      |
63 |            ▼                      |
64 |    ┌───────────────────────────┐  |
65 |    | Cleanup Temp Directories  |  |
66 |    | (Remove read-only attrs)  |  |
67 |    └───────────────────────────┘  |
68 └──────────────────────────────────┘
69             |
70             ▼
71     ✅ Complete: Code + PR + Docs
72
```